Transient Heat Conduction Time and Temperature Distributions in a Semi-Infinite Solid

Question

We shall examine the temperature in the semi-infinite solid in two ways. In the first way, the temperature is plotted over the range $0 \leq \eta \leq 5$ and $0.01 \leq \tau \leq 3$. In the second way, the temperature is plotted as a function of τ for $0.01 \leq \tau \leq 4$ at η = 0, 1, 2, 3, 4, and 5.

Solution

1. Governing Equation

$$\theta(x,t) = erfc\left(\frac{0.5}{t}\right) - e^{x+t^2} erfc\left(\frac{0.5x}{t} + t\right)$$

Were,

- $\eta = x$ (space similarity variable, non-dimensional)
- $\tau = t$ (time similarity variable, non-dimensional)
- θ = dimensionless temperature

2. Pick Values for Hand Calculation

Let's take a simple case:

$$x = 1.0,\ t = 0.5$$

3. Calculate First Term

$$A = \frac{0.5x}{t} = \frac{0.5 \cdot 1}{0.5} = 1.0$$

So,

$$erfc\,(A) = erfc(1.0)$$

From tables,
erf (1) = 0.8427
erfc (1) = 1 − 0.8427 = 0.1573

4. Calculate Second Term

$$B = \frac{0.5x}{t} + t = 1.0 + 0.5 = 1.5$$

So,
erf (1.5).
From tables,
erf (1.5) = 0.9661
erfc (1.5) = 1 − 0.9661 = 0.0339

Now compute the exponential part,

$$e^{x+t^2} = e^{1.0+(0.5)^2} = e^{1.25}$$
$$e^{1.25} \approx 3.4903$$

So, second term,

$$e^{x+t^2} \cdot erfc\,(B) = 3.4903 \cdot 0.0339 \approx 0.1183$$

5. Put Together

$$\theta(x,t) = erfc(1.0) - \ e^{1.25} \cdot erfc(1.5)$$
$$\theta(1{,}0.5) = 0.1573 - 0.1183 = 0.039$$

Result

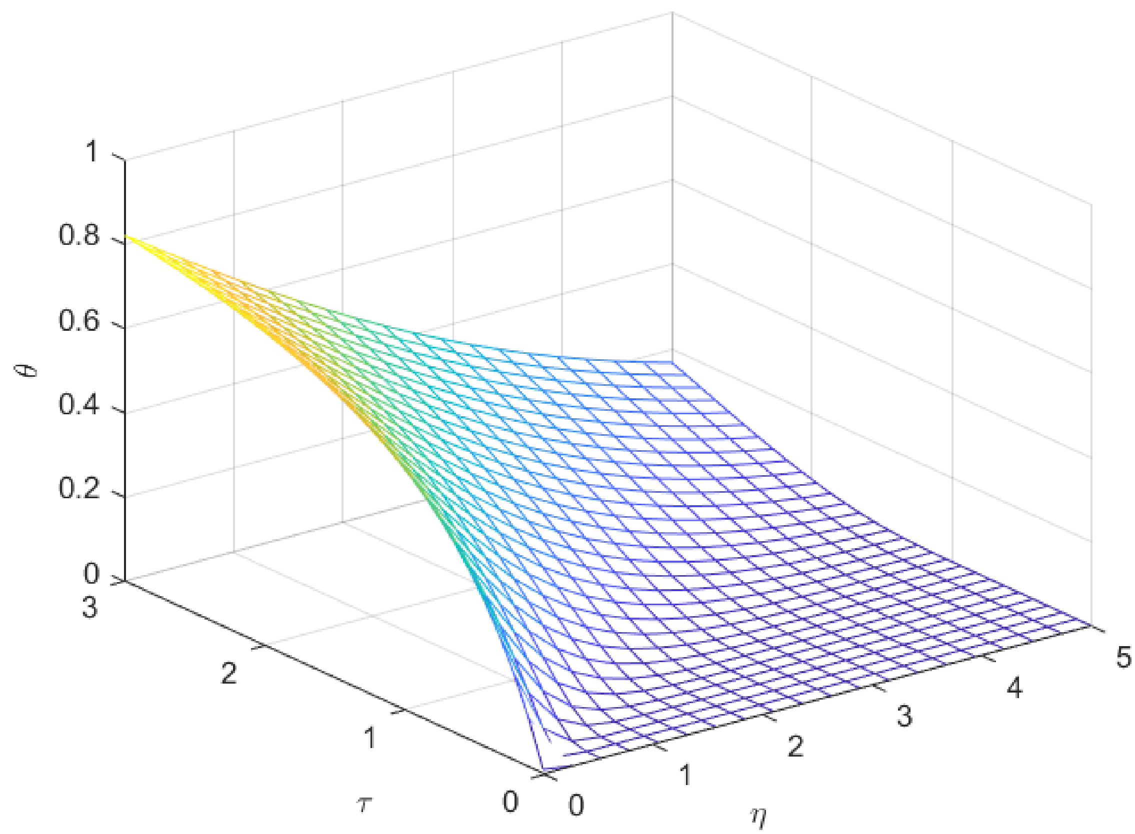

Figure 1 Temperature in a semi-infinite solid as a function of nondimensional position η and nondimensional time τ.

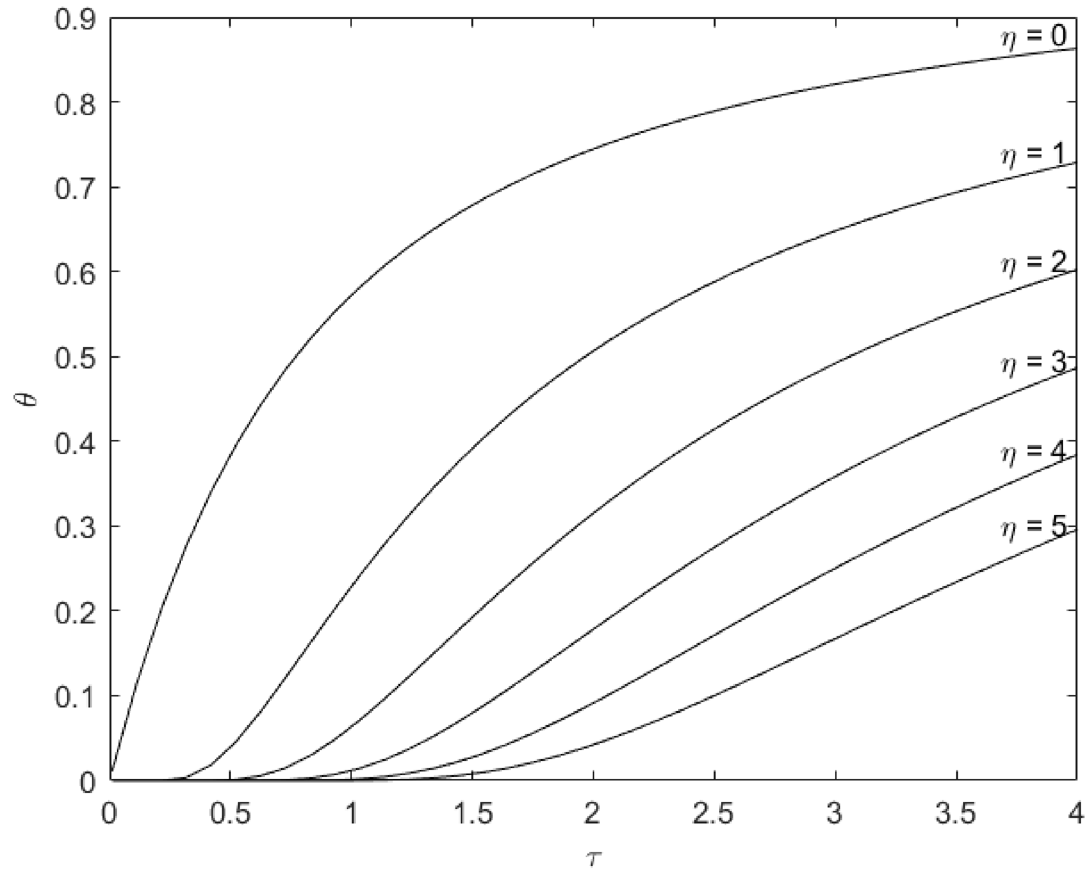

Figure 2 Temperature in a semi-infinite solid as a function of nondimensional time τ at several nondimensional locations η.